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2025 Group-II

Version No.				
3	2	0	8	1
0	0	●	0	0
1	1	1	1	●
2	●	2	2	2
●	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	●	8
9	9	9	9	9

ROLL NUMBER						
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1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

MATHEMATICS HSSC-I

SECTION – A (Marks 20)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اوّل لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Answer Sheet No. _____

Invigilator Sign. _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پُر کریں۔

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

	Question	A	B	C	D	A	B	C	D
1.	What are the multiplicative factors of z^2+9 ?	$(z+3i)(z-3i)$	$(z-3)(z+3)$	$(z+3)^2$	$(z-3i)^2$	○	○	○	○
2.	The rectangular form of $4\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$ is:	$-2+2i\sqrt{3}$	$-2-2i\sqrt{3}$	$2+2i\sqrt{3}$	$2-2i\sqrt{3}$	○	○	○	○
3.	A system of linear equations is said to be inconsistent if it has:	No solution	At least one solution	Exactly one solution	Exactly two solution	○	○	○	○
4.	If $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}=5$, then $\begin{vmatrix} 2a & 2b & 2c \\ 3d & 3e & 3f \\ g & h & i \end{vmatrix}=$	6	10	15	30	○	○	○	○
5.	In an arithmetic series, if $a=2$, $d=4$ then $S_{15}=$	240	450	480	510	○	○	○	○
6.	What is the sum of first two terms of an arithmetic-geometric sequence $T_n=(2n+1)\cdot 3^{n-1}$? $(n\in N)$	5	11	18	23	○	○	○	○
7.	For what value of x geometric series $1+2x+4x^2+8x^3+\ldots$ converges? $(r<1)$	$\frac{1}{2}$	$\frac{1}{4}$	1	$\frac{2}{1}$	○	○	○	○
8.	The number of ways to climb a 10-step staircase (taking either 1 or 2 steps at a time) is 89. Which method justifies the best conclusion?	Empirical counting	Pascal's Triangle	Mathematical Induction	Binomial Theorem	○	○	○	○
9.	What is the middle term in the expansion of $(x+y)^6$?	$\binom{6}{2}x^3y^3$	$\binom{6}{3}x^3y^3$	$\binom{6}{3}x^4y^2$	$\binom{6}{2}x^4y^2$	○	○	○	○
10.	The division of $2x^3-5x^2+3x-1$ by $(x-2)$ gives remainder:	–3	0	5	1	○	○	○	○
11.	If dot product of two vectors is zero then angle between them is:	180°	90°	45°	0°	○	○	○	○
12.	If $\vec{a}\perp\vec{b}$, then projection of $\vec{a}$ on $\vec{b}$ is:	–1	0	1	Undefined	○	○	○	○

	Question	A	B	C	D	A	B	C	D
13.	What is the volume of a parallelepiped formed by vectors $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$ and $\hat{k}+\hat{i}$?	0	1	$\sqrt{2}$	2	○	○	○	○
14.	$sin75^{\circ}cos15^{\circ} =$	$\frac{1}{2}(sin90^{\circ}+sin60^{\circ})$	$\frac{1}{2}(\cos 90^{\circ}+\cos 60^{\circ})$	$\frac{1}{2}(\sin 60^{\circ}-\sin 90^{\circ})$	$\frac{1}{2}(\cos 60^{\circ}-\cos 90^{\circ})$	○	○	○	○
15.	What is value of $cos\left(\frac{\theta}{2}\right)$, if $cos\theta=\frac{1}{2}$; $(0<\theta<90°)$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2\sqrt{2}}$	○	○	○	○
16.	Which of the following is an odd function?	$f(x)=cosx$	$f(x)=sinx+cosx$	$f(x)=tanx$	$f(x)=secx$	○	○	○	○
17.	The range of $f(x)=3cosx+4$ is:	$[-1, 1]$	$[1, 7]$	$[-3, 3]$	$[4, 7]$	○	○	○	○
18.	Which function's graph is symmetric about y-axis?	$y=sinx$	$y=tanx$	$y=cosx$	$y=cscx$	○	○	○	○
19.	A cell phone password consists of 4 distinct digits (0-9). How many possible passwords can be set?	${}^{10}C_4$	${}^{10}P_4$	${}^{9}C_4$	${}^{9}P_4$	○	○	○	○
20.	How many 5-letter words can be formed from the word 'TABLE' without repeating any letter?	20	60	120	240	○	○	○	○

——1HA-I 25008- 32081(D)——

$$S_n=\frac{n}{2}[2a+(n-1)d]$$

ROLL NUMBER						